

Tutorial 11: Hierarchical Clustering of Bird Assemblages

ENVX2001 – Applied Statistical Methods

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Semester 1

Dissimilarity Matrices and Hierarchical Clustering

In this tutorial we will use bird assemblage data from Mac Nally's forest and woodland bird study.

Mac Nally (1989) examined forest and woodland birds along a broad environmental gradient in south-eastern Australia. The paper focused on habitat breadth, habitat position and abundance, but the same dataset is also useful for exploring whether sites with similar bird assemblages cluster together.



Here, each row is a **site**, and each bird species column contains an abundance or density estimate for that species at that site.

We will use hierarchical clustering to ask:

Which sites have similar bird communities?

Load packages

We need the `vegan` package because it contains the `vegdist()` function for ecological dissimilarity matrices.

```
CODE  
library(vegan)
```

Load the data

The file `macnally.csv` contains the bird assemblage data.

If your file is in the same folder as this Quarto document use this :

```
CODE  
macnally <- read.csv("data/macnally.csv", row.names = 1)
```

Examine the structure of the data

```
CODE  
str(macnally)
```

```
OUTPUT  
'data.frame': 37 obs. of 103 variables:  
 $ HABITAT : chr "Mixed" "Gippsland Ma" "Gippsland Ma" "Gippsland Ma" ...  
 $ V16ST : num 3.4 3.4 8.4 3 5.6 8.1 8.3 4.6 3.2 4.6 ...  
 $ V2EYR : num 0 9.2 3.8 5 5.6 4.1 7.1 5.3 5.2 1.2 ...  
 $ V36F : num 0 0 0.7 0 12.9 10.9 6.9 11.1 8.3 4.6 ...  
 $ V4BTH : num 0 0 2.8 5 12.2 24.5 29.1 28.2 18.2 6.5 ...  
 $ V56WH : num 0 0 0 2 9.5 5.6 4.2 3.9 3.8 2.3 ...  
 $ V6WTTR : num 0 0 0 0 2.1 6.7 2 6.5 4.2 5.2 ...  
 $ V7WEHE : num 0 0 10.7 3 7.9 9.4 7.1 2.6 2.8 0.6 ...  
 $ V8WNHE : num 11.9 11.5 12.3 10 28.6 6.7 27.4 10.9 9 3.6 ...  
 $ V9SFW : num 0.4 8.3 4.9 6.9 9.2 0 13.1 3.1 3.8 3.8 ...  
 $ V10WBSW : num 0 12.6 10.7 12 5 8.9 2.8 8.6 5.6 3 ...  
 $ V11CR : num 1.1 0 0 0 19.1 12.1 0 9.3 14.1 7.5 ...  
 $ V12LK : num 3.8 0.5 1.9 2 3.6 6.7 2.8 3.8 3.2 2.4 ...  
 $ V13RWB : num 9.7 11.6 16.6 11 5.7 2.7 2.4 0.6 0 0.6 ...  
 $ V14AUR : num 0 0 2.3 1.5 8.8 0 2.8 1.3 0 0 ...  
 $ V15STTH : num 0 0 2.8 0 7 16.8 13.9 10.2 12.2 11.3 ...  
 $ V16LR : num 4.8 3.7 5.5 11 1.6 3.4 0 0 0.6 5.8 ...  
 $ V17WPHE : num 27.3 27.6 27.5 20 0 0 16.7 0 0 0 ...  
 $ V18YTH : num 0 0 0 0 0 0 0 0 9.6 ...
```

```

$ V19ER : num 5.1 2.7 5.3 2.1 1.4 2.2 0 1.2 1.3 2.3 ...
$ V20PCU : num 0 0 0 0 0 0 0 0 2.8 2.9 ...
$ V21ESP : num 0 3.7 0 2 3.5 3.4 5.5 5.1 7.1 0.6 ...
$ V22SCR : num 0 0 0 0 0.7 0 0 0 0 3 ...
$ V23RBFT : num 0 1.1 0 5 0 0.7 0 0.7 1.9 0 ...
$ V24BFCS : num 0.6 1.1 1.5 1.4 2.7 2 3.6 0 0.6 1.2 ...
$ V25WAG : num 1.9 3.4 2.1 3.4 0 0 0 0 0 0 ...
$ V26WWCH : num 0 0 0 0 0 0 0 0 9.8 ...
$ V27NHHE : num 0 6.9 3 32 6.4 2.2 5.6 0 0 0 ...
$ V28VS : num 0 0 0 0 0 5.4 5.6 1.9 4.2 5.1 ...
$ V29CST : num 1.7 0.9 1.5 1.4 0 0 4.6 0 0 0 ...
$ V30BTR : num 12.5 0 0 0 0 0 0 0 0 0 ...
$ V31AMAG : num 8.6 0 0 0 0 0 0 0 0 0 ...
$ V32SCC : num 12.5 0 0 0 0 0 0 0 0 0 ...
$ V33RWH : num 0.6 2.3 1.4 0 7 6.8 7.3 9.1 4.5 6.3 ...
$ V34WSW : num 0 5.7 24.3 10 0 0 3.6 0 0 0 ...
$ V35STP : num 4.8 0 3.1 4 0 0 2.4 0 0 3.5 ...
$ V36YFHE : num 0 1.1 11.7 0 6.1 0 0 0 0 2.3 ...
$ V37WHIP : num 0 0 0 0 0 0 0 2 3.2 0 ...
$ V38GAL : num 4.8 0 0 2.8 0 0 0 0 0 0 ...
$ V39FHE : num 26.2 0 0 0 0 0 0 0 0 0 ...
$ V40BRTH : num 0 0 0 0 0 0 0 0 0 6 ...
$ V41SPP : num 0 1.1 4.6 0.8 5.4 3.4 0 2 2.6 4.2 ...
$ V42SIL : num 0 0 0 0 0 2.7 1.2 0 4.9 1.2 ...
$ V43GCU : num 0 0 0 0 0 1.4 0 2.6 1.3 0 ...
$ V44MUSK : num 13.1 0 0 0 0 0 0 0 0 0 ...
$ V45MGLK : num 1.7 0 0 1.4 0 0 0 0 0 0 ...
$ V46BHHE : num 1.1 0 0 0 0 0 0 0 0 0 ...
$ V47RFC : num 0 0 0 0 0 0 0 0 0 0 ...
$ V48YTBC : num 0 0 0 0 0 0 0 1.9 2.6 0 ...
$ V49LYRE : num 0 0 0 0 0 0 0 0 0.6 0 ...
$ V50CHE : num 0 0 0 0 0 0 0 0 0.6 0 ...
$ V51OWH : num 0 0 0 0 0 0 0 0 0 0 ...
$ V52TRM : num 15 0 0 0 0 0 0 0 0 0 ...
$ V53MB : num 0 0 0 1 0 0 3.6 0 0 1.2 ...
$ V54STHR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V55LHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V56FTC : num 0 2.3 0 0 2.1 0 0 2.6 2.6 1.7 ...
$ V57PINK : num 0 0 0 0 0 0 0 0 0 0 ...
$ V58OBO : num 0 0 0 0 1.4 0 0 0 0 1.2 ...
$ V59YR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V60LFB : num 2.9 0 0 0 0 0 0 0 0 0 ...
$ V61SPW : num 0 0 0 0 0 0 0 0 0 0 ...
$ V62RBTR : num 0 0 0 0 0 0 0 0 0.6 ...
$ V63DWS : num 0.4 0 3.5 5.5 0 0 1.8 0 0 0 ...
$ V64BELL : num 0 0 0 0 22.1 0 0 0 0 0 ...
$ V65LWB : num 0 0 0 4 0 0 0 0 0 0 ...
$ V66CBW : num 0 0 0 0 0 0 0 0 0.6 ...
$ V67GGC : num 0 0 0 0 0 0 0 0 1.3 0 ...
$ V68PIL : num 0 0 0 0 0 0 0 0 1.3 0 ...
$ V69SKF : num 1.9 0 0 0 0 0 0 0 0 2.3 ...
$ V70RSL : num 6.7 0 0 0.8 0 0 0 0 0 0 ...
$ V71PD0V : num 0 0 0 0 0 0 0 0 0 0 ...
$ V72CRP : num 0 0 0 0 0 0 0 0 0 0 ...
$ V73JW : num 0 0 0 0 0 0 0 0 0 0 ...
$ V74BCHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V75RCR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V76GBB : num 0 0 0 0 0.7 0 0 0 0.6 0 ...
$ V77RRP : num 4.8 0 0 0 0 0 0 0 0 0 ...
$ V78LLOR : num 0 0 0 0 0 0 0 0 0 0 ...
$ V79YTHE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V80RF : num 0 0 0 0 0 1.4 0 1.9 3.2 0 ...
$ V81SHBC : num 0 0 1.4 0 0.7 0 0 2 1.3 0.6 ...
$ V82AZKF : num 0 0 0 0 0 0 0 0.7 0 0 ...
$ V83SFC : num 0 0 0 0 0 3.4 0 1.2 1.9 1.2 ...

```

```

$ V84YRTH : num 0 0 0 0 0 0 0 0 0 0 ...
$ V85ROSE : num 0 0 0 0 0 0 0 0.6 0 0 ...
$ V86BC00 : num 0 0 0 0 0 0 0 0.6 0 ...
$ V87LFC : num 0 0 1.8 0 0 0 2.4 0 0 0 ...
$ V88WG : num 0 0 0 0 0 0 0 0 0 0 ...
$ V89PC00 : num 1.9 0 0 0 0 0 0 0 0 0 ...
$ V90WTG : num 0 0 0 0 0 0 0 0 0 0 ...
$ V91NMIN : num 0.2 0 5.8 3.1 0 0 0 0 0 0 ...
$ V92NFB : num 0 0 0 0 0 0 0 0 0 0 ...
$ V93DB : num 0 0 0 0 0 0 0 0 0 0 ...
$ V94RBEE : num 0 0 0 0 0 0 0 0 0 0 ...
$ V95HBC : num 0 0 0 0 0 0 0 0 0 0 ...
$ V96DF : num 0 0 0 0 0 0 0 0 0 0 ...
$ V97PCL : num 9.1 0 0 0 0 0 0 0 0 0 ...
$ V98FLAME: num 0 0 0 0 0 0 0 0 0 0 ...
[1] output truncated

```

The first columns describe the site or habitat type. The remaining columns are bird species abundances.

```
CODE
ncol(macnally)
```

```
OUTPUT
[1] 103
```

```
CODE
nrow(macnally)
```

```
OUTPUT
[1] 37
```

Separate site information from species data

The HABITAT column describes the broad habitat category. The bird species data begin after the habitat column.

```
CODE
site_info <- macnally[, 1, drop = FALSE]
species_data <- macnally[, 2:ncol(macnally)]

head(site_info)
```

```
OUTPUT
```

	HABITAT
Reedy Lake	Mixed
Pearcedale	Gippsland Ma
Warneet	Gippsland Ma
Cranbourne	Gippsland Ma
Lysterfield	Mixed
Red Hill	Mixed

CODE

```
head(species_data[, 1:6])
```

OUTPUT

	V16ST	V2EYR	V3GF	V4BTH	V5GWH	V6WTTR
Reedy Lake	3.4	0.0	0.0	0.0	0.0	0.0
Pearcedale	3.4	9.2	0.0	0.0	0.0	0.0
Warneet	8.4	3.8	0.7	2.8	0.0	0.0
Cranbourne	3.0	5.0	0.0	5.0	2.0	0.0
Lysterfield	5.6	5.6	12.9	12.2	9.5	2.1
Red Hill	8.1	4.1	10.9	24.5	5.6	6.7

Check that the species data are numeric:

CODE

```
species_data <- as.data.frame(lapply(species_data, as.numeric))  
rownames(species_data) <- rownames(macnally)
```

Bray-Curtis Dissimilarity

Calculate Bray-Curtis dissimilarity

Bray-Curtis dissimilarity is commonly used for ecological community data because it compares sites using species abundances.

Values range from:

- 0 = sites have identical assemblages
- 1 = sites share no species in common

CODE

```
Braydistance <- vegdist(species_data, method = "bray")  
Braydistance
```

OUTPUT

	Reedy Lake	Pearcedale	Warneet	Cranbourne	Lysterfield	Red Hill
Pearcedale	0.6064757					
Warneet	0.6192469	0.3608724				
Cranbourne	0.6493576	0.3458135	0.3881690			
Lysterfield	0.8504073	0.6274208	0.6067270	0.6679905		
Red Hill	0.8646783	0.6974849	0.6395924	0.7012250	0.4194429	
Devilbend	0.7803703	0.5554849	0.5432526	0.5666394	0.4136546	0.4006753
Olinda	0.8728324	0.6716754	0.6963958	0.6993095	0.4338760	0.2425920
Fern Tree Gu	0.8842927	0.6968785	0.7297463	0.7147345	0.4499714	0.2855280
Sherwin	0.8476881	0.7610514	0.7041306	0.7377265	0.5725373	0.4657534
Heathcote Ju	0.8974209	0.7467483	0.6844389	0.7278521	0.5267883	0.3536222
Warburton	0.8153215	0.6633222	0.6380952	0.6420233	0.4095324	0.4015512

Millgrove	0.8219493	0.6747027	0.6293312	0.6742779	0.4030380	0.2875000
Ben Cairn	0.9311377	0.7501558	0.7619048	0.7675415	0.4907385	0.3561757
Panton Gap	0.9123711	0.7683323	0.7341635	0.7415419	0.5327723	0.3789745
OShannassy	0.9066493	0.7286796	0.6514113	0.7006459	0.5036120	0.3789901
Ghin Ghin	0.7792014	0.7436293	0.6975762	0.7337265	0.6096528	0.6017699
Minto	0.7689873	0.7439673	0.6728462	0.7266347	0.4920580	0.5161351
Hawke	0.8759055	0.6424064	0.6380060	0.6511548	0.4881841	0.3862168
St Andrews	0.8248432	0.7740822	0.7041327	0.7450822	0.4629906	0.3982202
Nepean	0.8774592	0.6449275	0.6055753	0.6705238	0.4548486	0.2912591
Cape Schanck	0.8097156	0.5754331	0.6039081	0.4468775	0.5386148	0.5345982
Balnarring	0.8849534	0.6780521	0.6940319	0.6294168	0.5442571	0.4557338
Bittern	0.7009615	0.4164188	0.3341721	0.4724653	0.5248750	0.6027579
Bailleston	0.8789157	0.8065028	0.7153797	0.7820092	0.5769116	0.5175708
Donna Buang	0.9324841	0.8249360	0.7481381	0.7836706	0.5963533	0.4998182
Upper Yarra	0.8625863	0.7412229	0.6847792	0.6992481	0.4605809	0.3933424
Gembrook	0.8512503	0.6490595	0.6087247	0.6867196	0.4669492	0.4233100
Arcadia	0.5150546	0.6724196	0.6770395	0.6935901	0.8977333	0.8852556
Undera	0.6664769	0.7503682	0.7536058	0.7271605	0.8289733	0.7920000
Coomboona	0.5987567	0.7854798	0.7184611	0.7378254	0.8642458	0.8349206
Toolamba	0.5382810	0.7081578	0.6773193	0.6555024	0.8985507	0.9092715
Rushworth	0.7271251	0.7596685	0.6550741	0.7736842	0.7194110	0.7228442
Sayers	0.7972973	0.7429010	0.6656308	0.7265176	0.5889695	0.6132670
Waranga	0.4585441	0.5790878	0.4946113	0.5544114	0.7367044	0.8186628
Costerfield	0.7079971	0.7520510	0.6940815	0.7289646	0.5651332	0.5508108
Tallarook	0.8204724	0.7350598	0.6655629	0.7013575	0.5102571	0.4989769
Devilbend	0.4134984	0.2191262	0.4477504	0.3338252	0.4297782	0.3087515
Olanda	0.4766962	0.3420229	0.3545526	0.5663501	0.3953079	0.3535584
Fern Tree Gu	0.6033313	0.2948987	0.3417279	0.3338252	0.3577087	0.2946429
Sherwin	0.5007474	0.4322760	0.4322760	0.3420229	0.3545526	0.5663501
Heathcote Ju	0.4322760	0.2527538	0.2628129	0.4519950	0.3440048	0.3087515
Warburton	0.4336283	0.5595027	0.3204164	0.2845152	0.5827034	0.3953079
Millgrove	0.5174403	0.3213213	0.3285242	0.5570934	0.3577087	0.2946429
Ben Cairn	0.5095541	0.3168950	0.3369896	0.5488194	0.3524869	0.3564474
Panton Gap	0.6204506	0.6060973	0.5825666	0.5971873	0.5926561	0.6212690
OShannassy	0.4673546	0.5371740	0.5696203	0.6895929	0.5626719	0.4924142
Ghin Ghin	0.5231660	0.3600000	0.3729977	0.3831851	0.2821110	0.4626866
Minto	0.4732006	0.3973799	0.4644522	0.3684531	0.3327428	0.4698464
Hawke	0.4185047	0.3304521	0.3450915	0.4406720	0.3432432	0.4372140
St Andrews	0.6057348	0.5173824	0.5233918	0.6619804	0.5887407	0.5603587
Nepean	0.4761388	0.4845554	0.4672080	0.5907007	0.5400697	0.4703462
Cape Schanck	0.4634742	0.6365651	0.6201830	0.7380746	0.6490736	0.5515873
Balnarring	0.5800799	0.5530686	0.5880014	0.3974800	0.4930198	0.6287879
Bittern	0.6687276	0.4911197	0.4155392	0.6121361	0.5444532	0.4761092
Bailleston	0.5284752	0.3381243	0.3424460	0.4422492	0.2936015	0.4375843
Donna Buang	0.4960971	0.4274643	0.4357731	0.5726496	0.4883607	0.4144411
Upper Yarra	0.7742811	0.9157040	0.8675000	0.7673657	0.8104596	0.8658271
Gembrook	0.7156948	0.8029690	0.8007430	0.5982936	0.7263374	0.7974293
Arcadia	0.7540323	0.8527936	0.8439384	0.6920366	0.7973478	0.8528736
Undera	0.7833633	0.9223505	0.9215686	0.8183161	0.8641053	0.9059445
Coomboona	0.7772264	0.7372774	0.7102833	0.5115269	0.6220930	0.6673219
Toolamba	0.6973329	0.6306938	0.6050509	0.4228635	0.5210264	0.5603175
Rushworth	0.7004464	0.8108581	0.8127191	0.7171087	0.7599392	0.6790575
Sayers	0.6066634	0.5512857	0.5486425	0.3984652	0.4924091	0.5216125
Waranga	0.5442159	0.4349693	0.4805035	0.4377613	0.4769614	0.4476828
Costerfield	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Tallarook	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Millgrove	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Ben Cairn	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Panton Gap	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
OShannassy	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Ghin Ghin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Minto	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Heathcote Ju	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Warburton	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Devilbend	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Olanda	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Fern Tree Gu	0.4769614	0.4476828	0.4769614	0.4476828	0.4769614	0.4476828
Sherwin	0.4769614	0.4476828	0.4769614	0.4476828		

Cranbourne						
Lysterfield						
Red Hill						
Devilbend						
Olinda						
Fern Tree Gu						
Sherwin						
Heathcote Ju						
Warburton						
Millgrove						
Ben Cairn	0.3816227					
Panton Gap	0.3855517	0.2769469				
OShannassy	0.3444028	0.3368119	0.2825911			
Ghin Ghin	0.5582715	0.6406751	0.6173951	0.5847382		
Minto	0.5038482	0.5402744	0.5465995	0.5135135	0.4359377	
Hawke	0.4007807	0.4667507	0.4292282	0.4006667	0.6044699	0.6138471
St Andrews	0.3769310	0.5013809	0.4355383	0.4902887	0.6200086	0.5528872
Nepean	0.3439398	0.3925759	0.3827305	0.3727461	0.5915859	0.5844156
Cape Schanck	0.4974174	0.5930496	0.5732501	0.5413375	0.6801550	0.6878429
Balnarring	0.4707170	0.5241140	0.5177931	0.5536313	0.5880759	0.6337199
Bittern	0.5501355	0.6488696	0.6377171	0.6105395	0.5832816	0.5474478
Bailieston	0.5563840	0.6859554	0.6163009	0.6216216	0.7049837	0.7226891
Donna Buang	0.5044423	0.4251115	0.4385382	0.4623946	0.6957776	0.7023661
Upper Yarra	0.3056931	0.4362053	0.3818295	0.3122229	0.5676485	0.5537817
Gembrook	0.3818261	0.4676074	0.4231699	0.4742451	0.6147235	0.5164386
Arcadia	0.8406353	0.8512397	0.8752415	0.8753494	0.6615454	0.7160744
Undera	0.7779080	0.8388383	0.8280142	0.8191075	0.6006687	0.7253467
Coomboona	0.8022106	0.8857789	0.8703432	0.8561237	0.5982906	0.6940771
Toolamba	0.8736397	0.9242736	0.9097286	0.9084725	0.6477217	0.7306861
Rushworth	0.6482048	0.8109426	0.7065095	0.7021881	0.7348294	0.7706627
Sayers	0.5700878	0.6919459	0.6096518	0.6141220	0.6590630	0.7217631
Waranga	0.7237967	0.8701870	0.7808467	0.7616651	0.7061303	0.6873896
Costerfield	0.5090271	0.6535356	0.6079818	0.5891671	0.6383872	0.6715797
Tallarook	0.3885144	0.5861345	0.5385870	0.5510316	0.6368503	0.6410350
	Hawke	St Andrews	Nepean	Cape Schanck	Balnarring	Bittern
Pearcedale						
Warneet						
Cranbourne						
Lysterfield						
Red Hill						
Devilbend						
Olinda						
Fern Tree Gu						
Sherwin						
Heathcote Ju						
Warburton						
Millgrove						
Ben Cairn						
Panton Gap						
OShannassy						
Ghin Ghin						
Minto						
Hawke						
St Andrews	0.4019048					
Nepean	0.3250092	0.4384376				
Cape Schanck	0.5188018	0.6144883	0.4728770			
Balnarring	0.5020548	0.6064343	0.4126552	0.4636902		
Bittern	0.6282707	0.6575574	0.5830636	0.5561484	0.5528809	
Bailieston	0.5122736	0.4427921	0.4783821	0.7228145	0.6600326	0.6812680
Donna Buang	0.5062907	0.6192616	0.5238095	0.6848211	0.6693241	0.6990881
Upper Yarra	0.3290633	0.4195889	0.4015967	0.5533981	0.5282651	0.5963250
Gembrook	0.5176899	0.4873227	0.3702389	0.4872900	0.4995479	0.4972044
Arcadia	0.8581220	0.8015021	0.8727658	0.9031216	0.8250429	0.6776838
Undera	0.7334826	0.7128117	0.7387329	0.8740293	0.8073029	0.7920306
Coomboona	0.8269231	0.7660785	0.8082281	0.8852085	0.8404385	0.8124153

Toolamba	0.8811189	0.8617021	0.8823132	0.9039146	0.8909321	0.7617308
Rushworth	0.6383351	0.5819024	0.6711906	0.6844408	0.7441117	0.6524454
Sayers	0.4573764	0.4613791	0.5392954	0.6804277	0.6760876	0.6696680
Waranga	0.7580819	0.7262273	0.7748474	0.6913525	0.7921715	0.5439642
Costerfield	0.5119852	0.4434055	0.4995876	0.6648569	0.6510172	0.6551073
Tallarook	0.4063866	0.3479524	0.4740917	0.6066790	0.5504923	0.6338384
Bailieston	Donna Buang	Upper Yarra	Gembrook	Arcadia	Undera	
Pearcedale						
Warneet						
Cranbourne						
Lysterfield						
Red Hill						
Devilbend						
Olinda						
Fern Tree Gu						
Sherwin						
Heathcote Ju						
Warburton						
Millgrove						
Ben Cairn						
Panton Gap						
OShannassy						
Ghin Ghin						
Minto						
Hawke						
St Andrews						
Nepean						
Cape Schanck						
Balnarring						
Bittern						
Bailieston						
Donna Buang	0.6938776					
Upper Yarra	0.5406735	0.4819326				
Gembrook	0.5788418	0.6313993	0.4852574			
Arcadia	0.8367613	0.8640305	0.8322497	0.8923564		
Undera	0.6692635	0.8449319	0.7603102	0.8135310	0.4860301	
Coomboona	0.7507898	0.8546272	0.7945493	0.8650190	0.3845216	0.3125205
Toolamba	0.8661273	0.9251674	0.8726208	0.9156449	0.3077411	0.4637830
Rushworth	0.4846871	0.8059490	0.6778257	0.6053174	0.8359034	0.7135710
Sayers	0.4089904	0.6835043	0.5870920	0.5781979	0.8322212	0.6650619
Waranga	0.7054706	0.8874082	0.7565891	0.6848160	0.6130536	0.6573686
Costerfield	0.3962930	0.6626281	0.5293083	0.5338377	0.7870564	0.6370721
Tallarook	0.5121795	0.6394558	0.4471752	0.4963161	0.8609626	0.7549789
Coomboona	Toolamba	Rushworth	Sayers	Waranga	Costerfield	
Pearcedale						
Warneet						
Cranbourne						
Lysterfield						
Red Hill						
Devilbend						
Olinda						
Fern Tree Gu						
Sherwin						
Heathcote Ju						
Warburton						
Millgrove						
Ben Cairn						
Panton Gap						
OShannassy						
Ghin Ghin						
Minto						
Hawke						
St Andrews						
Nepean						
Cape Schanck						

```
Balnarring
Bittern
Baillieston
Donna Buang
Upper Yarra
Gembrook
Arcadia
Undera
Coomboona
Toolamba      0.2675289
Rushworth     0.7728824 0.8609764
Sayers        0.7899761 0.8572958 0.3555699
Waranga       0.6746811 0.6315256 0.4422122 0.6249683
Costerfield   0.7307905 0.8239679 0.3983567 0.3017514 0.5793342
Tallarook     0.7831686 0.8734808 0.5792788 0.4818976 0.6899011 0.4121882
```

Hierarchical Clustering

Run hierarchical clustering

We will use UPGMA clustering, which is specified by `method = "average"` in `hclust()`.

CODE

```
hc ← hclust(Braydistance, method = "average")
hc
```

OUTPUT

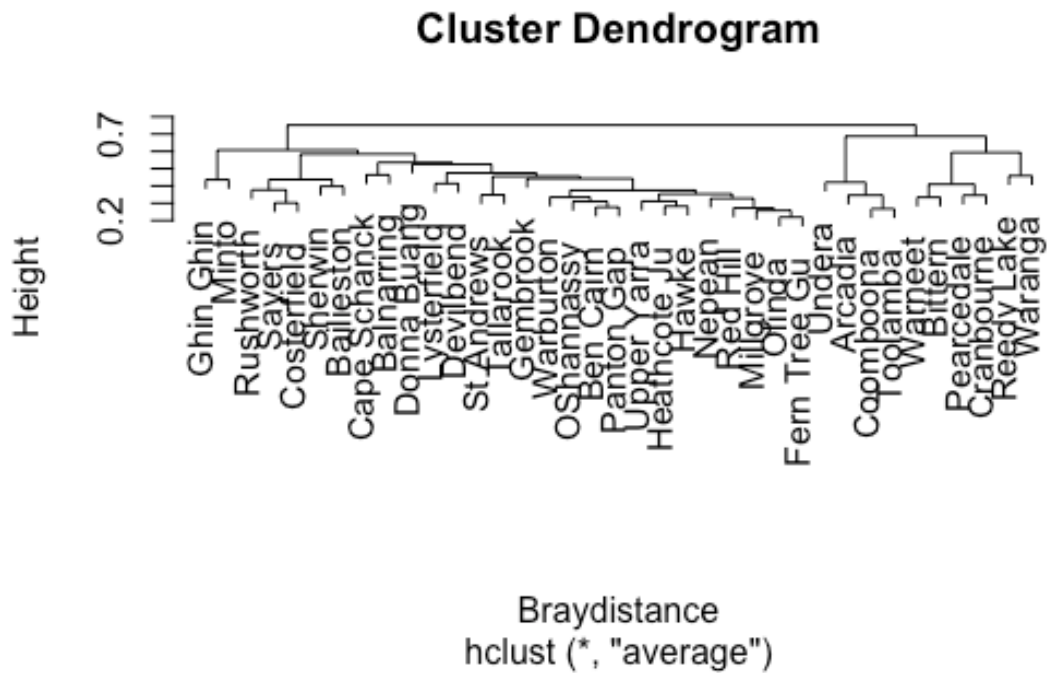
```
Call:
hclust(d = Braydistance, method = "average")

Cluster method : average
Distance       : bray
Number of objects: 37
```

Plot the dendrogram

CODE

```
plot(hc)
```



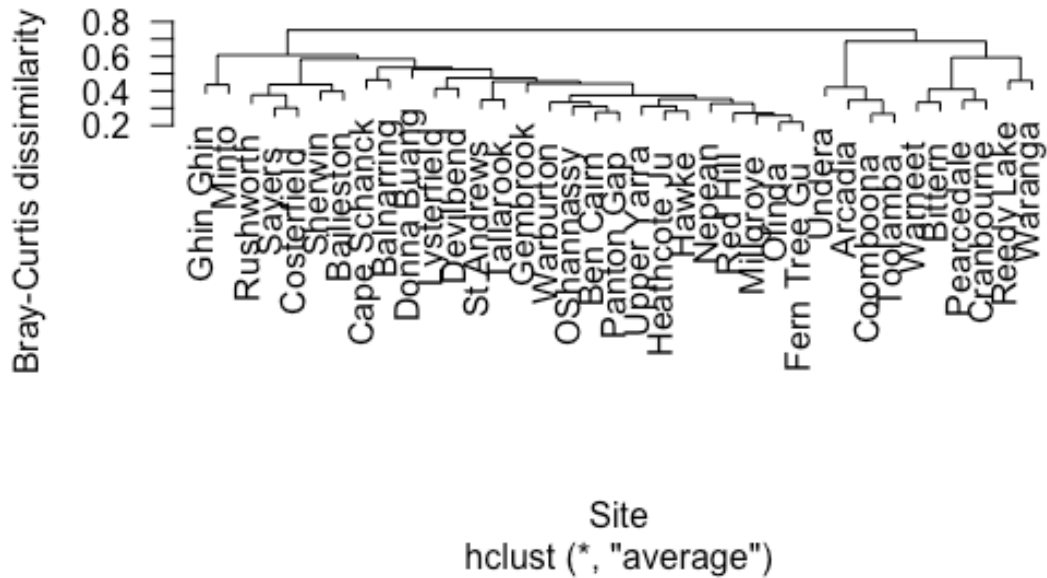
The dendrogram shows how sites are progressively joined into clusters.

Sites joined low on the dendrogram are more similar. Sites joined high on the dendrogram are more different.

Improved dendrogram with labels

```
CODE
plot(hc,
  las = 1,
  main = "Hierarchical clustering of bird assemblages",
  xlab = "Site",
  ylab = "Bray-Curtis dissimilarity")
```

Hierarchical clustering of bird assemblages



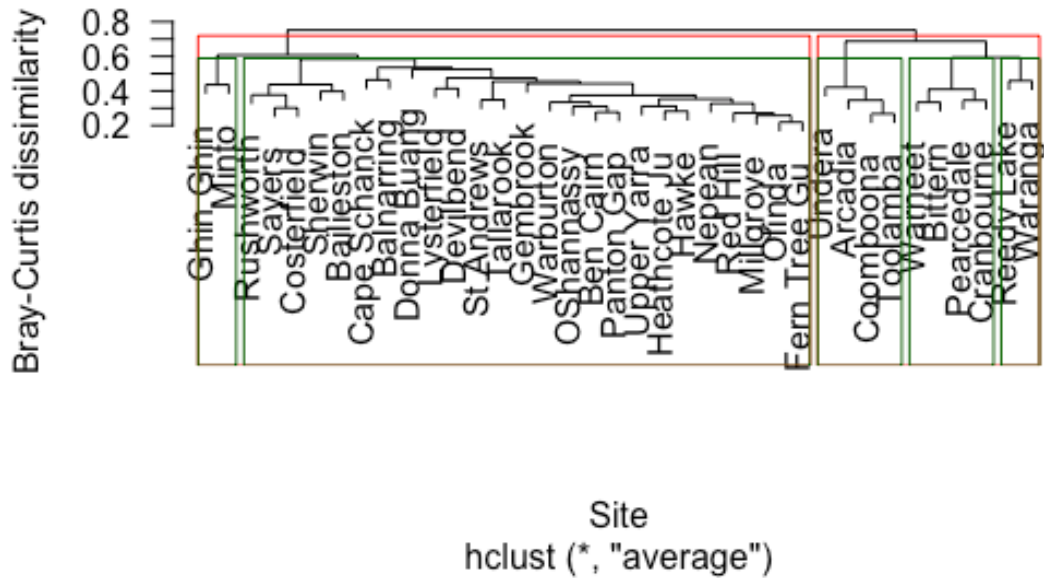
Add cluster rectangles

We can draw rectangles around possible cluster solutions.

```
CODE
plot(hc,
     las = 1,
     main = "Cluster diagram of bird assemblages",
     xlab = "Site",
     ylab = "Bray-Curtis dissimilarity")

rect.hclust(hc, k = 2, border = "red")
rect.hclust(hc, k = 5, border = "darkgreen")
```

Cluster diagram of bird assemblages



The rectangles show possible groupings:

- **Red rectangles** = 2 broad clusters
- **Green rectangles** = 5 finer clusters

There is no single correct number of clusters. The appropriate choice depends on the ecological question.

Extracting Cluster Membership

Cut the tree into clusters

Cut the tree into 5 clusters:

```
CODE
clusters5 <- cutree(hc, k = 5)
clusters5
```

OUTPUT

Reedy Lake	Pearcedale	Warneet	Cranbourne	Lysterfield	Red Hill
1	2	2	2	3	3
Devilbend	Olinda	Fern Tree Gu	Sherwin	Heathcote Ju	Warburton
3	3	3	3	3	3
Millgrove	Ben Cairn	Panton Gap	OShannassy	Ghin Ghin	Minto
3	3	3	3	4	4
Hawke	St Andrews	Nepean	Cape Schanck	Balnarring	Bittern
3	3	3	3	3	2
Bailieston	Donna Buang	Upper Yarra	Gembrook	Arcadia	Undera
3	3	3	3	5	5
Coomboona	Toolamba	Rushworth	Sayers	Waranga	Costerfield
5	5	3	3	1	3
Tallarook					
3					

Add the cluster identity back to the site information:

CODE

```
cluster_summary <- data.frame(
  Site = rownames(macnally),
  Habitat = macnally$HABITAT,
  Cluster = clusters5
)
```

cluster_summary

OUTPUT

	Site	Habitat	Cluster
Reedy Lake	Reedy Lake	Mixed	1
Pearcedale	Pearcedale	Gippsland Ma	2
Warneet	Warneet	Gippsland Ma	2
Cranbourne	Cranbourne	Gippsland Ma	2
Lysterfield	Lysterfield	Mixed	3
Red Hill	Red Hill	Mixed	3
Devilbend	Devilbend	Mixed	3
Olinda	Olinda	Mixed	3
Fern Tree Gu	Fern Tree Gu	Montane Fore	3
Sherwin	Sherwin	Foothills Wo	3
Heathcote Ju	Heathcote Ju	Montane Fore	3
Warburton	Warburton	Montane Fore	3
Millgrove	Millgrove	Mixed	3
Ben Cairn	Ben Cairn	Mixed	3
Panton Gap	Panton Gap	Montane Fore	3
OShannassy	OShannassy	Mixed	3
Ghin Ghin	Ghin Ghin	Mixed	4
Minto	Minto	Mixed	4
Hawke	Hawke	Mixed	3
St Andrews	St Andrews	Foothills Wo	3
Nepean	Nepean	Foothills Wo	3
Cape Schanck	Cape Schanck	Mixed	3
Balnarring	Balnarring	Mixed	3
Bittern	Bittern	Gippsland Ma	2
Bailieston	Bailieston	Box-Ironbark	3
Donna Buang	Donna Buang	Mixed	3
Upper Yarra	Upper Yarra	Mixed	3
Gembrook	Gembrook	Mixed	3
Arcadia	Arcadia	River Red Gu	5
Undera	Undera	River Red Gu	5
Coomboona	Coomboona	River Red Gu	5
Toolamba	Toolamba	River Red Gu	5
Rushworth	Rushworth	Box-Ironbark	3
Sayers	Sayers	Box-Ironbark	3

Waranga	Waranga	Mixed	1
Costerfield	Costerfield	Box-Ironbark	3
Tallaroook	Tallaroook	Foothills Wo	3

How many sites are in each cluster?

```
CODE
table(cluster_summary$Cluster)
```

```
OUTPUT

 1  2  3  4  5
2  4 25  2  4
```

Do clusters correspond to habitat types?

```
CODE
table(cluster_summary$Habitat, cluster_summary$Cluster)
```

```
OUTPUT

      1  2  3  4  5
Box-Ironbark 0  0  4  0  0
Foothills Wo 0  0  4  0  0
Gippsland Ma 0  4  0  0  0
Mixed        2  0 13  2  0
Montane Fore 0  0  4  0  0
River Red Gu 0  0  0  0  4
```

This table helps us ask whether bird assemblage clusters correspond to habitat categories.

If sites from the same habitat type mostly fall into the same cluster, then bird communities are strongly structured by habitat.

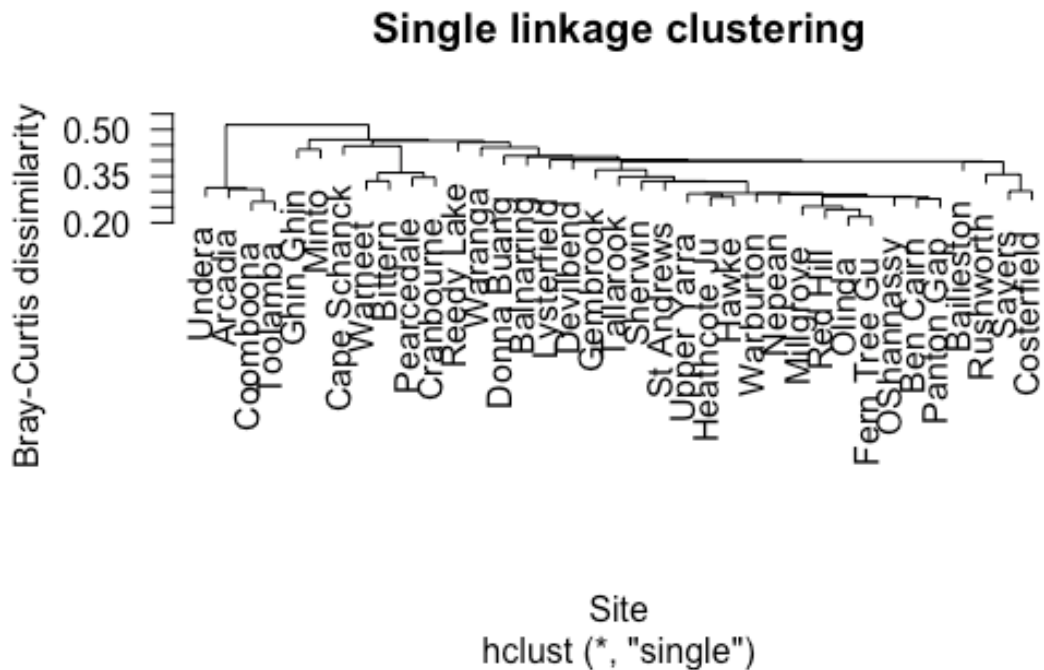
If habitat types are spread across several clusters, then bird assemblages may be influenced by other factors as well.

Optional: Compare Clustering Methods

Single linkage

```
CODE
hc_single <- hclust(Braydistance, method = "single")

plot(hc_single,
     las = 1,
     main = "Single linkage clustering",
     xlab = "Site",
     ylab = "Bray-Curtis dissimilarity")
```



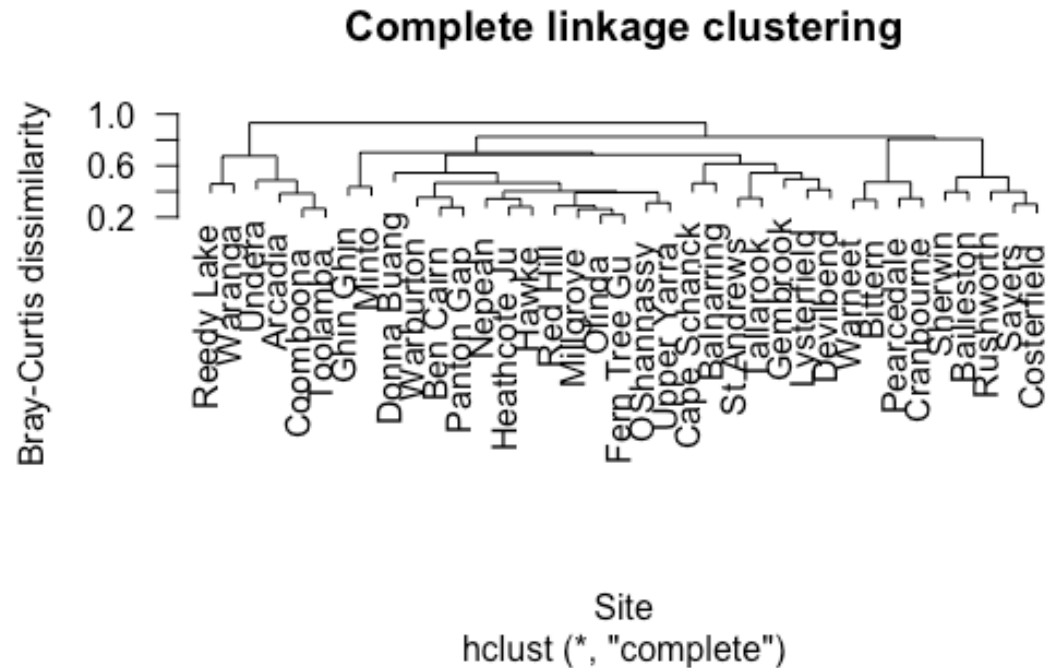
Single linkage can produce chaining, where sites are joined one at a time into long strings.

Complete linkage

```
CODE
hc_complete <- hclust(Braydistance, method = "complete")

plot(hc_complete,
     las = 1,
     main = "Complete linkage clustering",
```

```
xlab = "Site",
ylab = "Bray-Curtis dissimilarity")
```

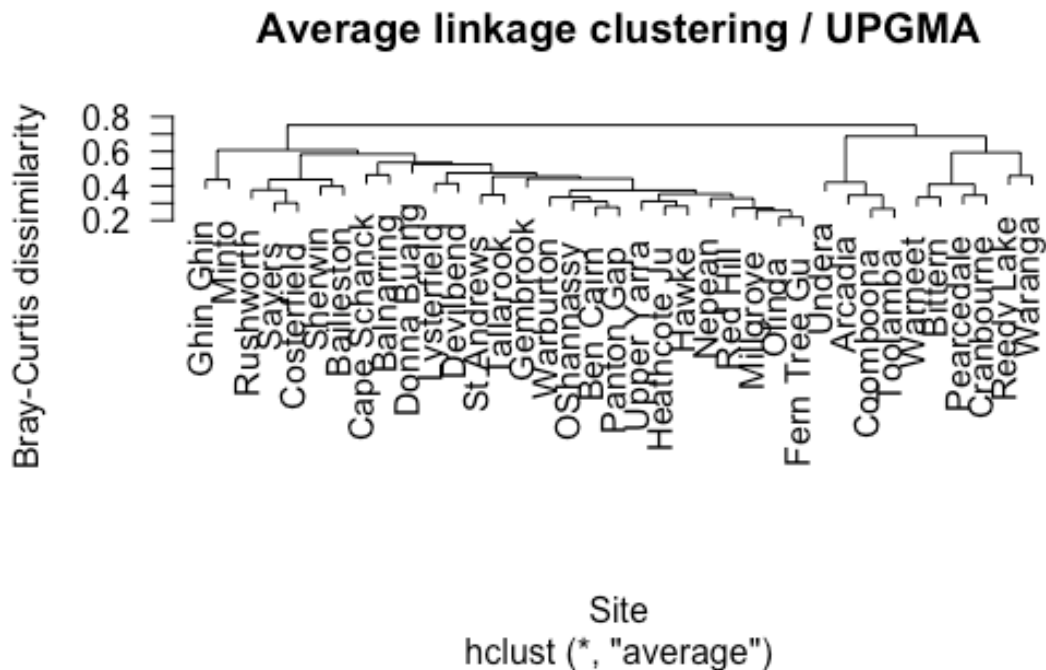


Complete linkage tends to produce more compact clusters.

Average linkage

```
CODE
hc_average <- hclust(Braydistance, method = "average")

plot(hc_average,
     las = 1,
     main = "Average linkage clustering / UPGMA",
     xlab = "Site",
     ylab = "Bray-Curtis dissimilarity")
```



Average linkage is often a useful compromise and is commonly used in ecological community analysis.

Student Questions

Question 1

What does Bray-Curtis dissimilarity measure in this analysis?

Answer guide

It measures how different two sites are in their bird assemblages, based on species abundances.

A value near 0 means sites are very similar. A value near 1 means sites are very different.

Question 2

What does it mean when two sites join low on the dendrogram?

Answer guide

They have similar bird assemblages.

The lower the joining point, the smaller the dissimilarity between the sites.

Question 3

Why might we compare 2-cluster and 5-cluster solutions?

Answer guide

A 2-cluster solution shows very broad differences among sites.

A 5-cluster solution shows finer ecological structure.

The best level depends on whether we want a broad or detailed interpretation.

Question 4

Why might clusters not perfectly match habitat categories?

Answer guide

Bird assemblages may respond to more than broad habitat type.

They may also be influenced by vegetation structure, geographic location, resource availability, disturbance, or species interactions.

Key Take-home Message

Hierarchical clustering groups sites according to similarity in bird assemblages.

For the Mac Nally bird data, this lets us explore whether woodland and forest sites form distinct community types.

The method does not tell us the “true” number of ecological groups. Instead, it gives a visual and quantitative way to explore patterns in community composition.